



GE Medical Systems

Technical Publications

Direction 2316316

Revision 4

TwinSpeed Interface Ring Alignment

Copyright© 2002 by General Electric Co.

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

Rev. 11/15/2000

Direction 2128126 - Language Policy For Service Documentation, Rev 0

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルは 英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
0	Nov. 26, 2001	Initial version of TwinSpeed Alignment Procedure.
1	Feb 18, 2002	Added magnet flange check and procedures to address issues with magnet flange.
2	March 30, 2002	Modified procedure
3	Aug 2, 2002	Changed part numbers for o-rings due to change
4	Jan 14, 2003	Changed correct part number for Upgrade kit

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page.....	0	Title Page	1	Title Page	2	Title Page.....	3
Damage in Trans.	-	Damage in Trans.	-	Damage in Trans....	-	Damage in Trans.	-
Direction 2128126 .0*		Direction 2128126. 0*		Direction 2128126 .0*		Direction 2128126..0*	
6 to 32	0	6 to 32	1	6 to 37	2	6 to 37	3

PAGE	REV
Title Page.....	4
Damage in Trans.	-
Direction 2128126 .0*	
6 to 37	4

* This revision/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

1- OVERVIEW	7
2- REQUIRED TOOLS & EQUIPMENT	7
3- MAGNET FLANGE CHECK.....	8
4- TOOL SETUP.....	9
5- PREPARATION FOR INTERFACE RING.....	12
6- ALIGNMENT PROCEDURE	17
6-1 Rotational Alignment.....	18
6-2 X-Y Plane Concentric Alignment	19
6-3 X & Y-Axis Alignment.....	22
6-3-1 Y-Axis Alignment	22
6-3-2 X-Axis Alignment	23
6-4 Securing Adjustments.....	23
6-5 Rear Interface Ring Alignment.....	24
APPENDIX A – DIGITAL PROTRACTOR TESTING.....	25
A-1 Recalibration	25
APPENDIX B – TWIN CENTERING TOOL KIT	30
APPENDIX C – MAGNET FLANGE ISSUES.....	33
C-1 Protruding Magnet Weld	33
C-2 Cavity in Magnet Flange	35
C-3 Miss-aligned Hole in Magnet Flange.....	36
C4 Skewed Magnet Flange.....	37

1- OVERVIEW

A new tool is required to align the TwinSpeed Vacuum Enclosure Interface Rings to each other for TwinSpeed upgrades (new systems are factory aligned). The vacuum end bells are mounted to the interface rings and then the RF coil is mounted to the end bells. The alignment of the interface rings is critical to the performance of the vacuum enclosure. The RF coil alignment to the gradient coils is critical to SNR performance of the system. Since the RF coil in a TwinSpeed system is mounted/supported by the enclosure end bells, the alignment of the RF coil is ultimately determined by the interface ring alignment.

2- REQUIRED TOOLS & EQUIPMENT

The tools listed below that are highlighted in gray are the parts required for the Magnet Interface Rings. The remaining parts are required for the Gradient Isolation Brackets. These parts are needed after the coil is installed in the magnet.

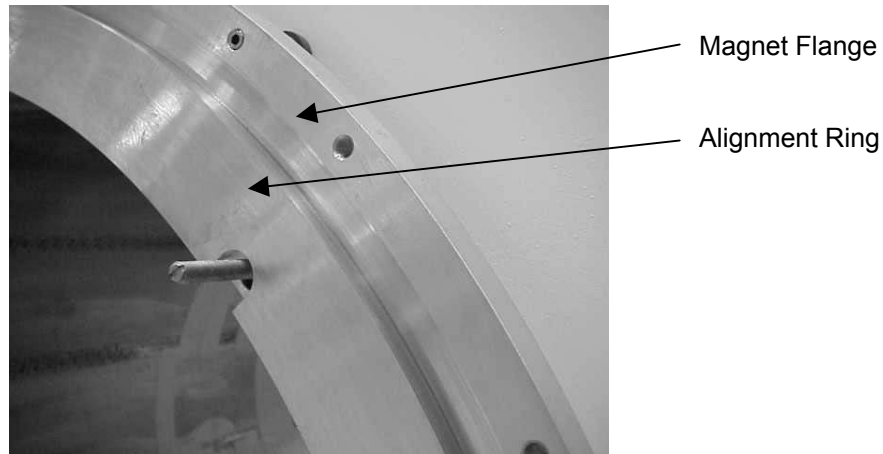
Description	Part Number	Qty.
<u>Twin Centering Tool Kit</u> (Order from Tool Depot) (Kit weighs approx. 78 Lbs./ 39 kg)	2299496	1
<u>TRM LCC Gradient ISO Install Kit</u>	2307415	1
SUPPORT MOUNTING, TRM	2272016	2
SPACER SHIM 1.5MM THK	2238908	2
SPACER SHIM 2.4MM THK	2238908-2	6
SPACER SHIM 16MM THK	2238908-3	2
RUBBER ISOLATION PAD	2241176	2
SCREW SET SCREW 10 MM 25 MM--Grub	2251295	2
SCREW SET SCREW 10 MM 30 MM--Grub	2251295-2	1
MTG PAD, 1/4" THICK NEOPRENE	2131953-5	4
SET SCREW- MTG.PAD	2293535	4
RADIAL SHIM SPACER 1.5MM THK	2293771	4
RADIAL SHIM SPACER 0.8MM THK	2293771-2	8
MAGNET RING	2306226	2
WEDGE RING	2306184	2
O-RING WEDGE RING	2310035	2
MAGNET RING GASKET	2246966	2
SCREW HEXAGON SOCKET 10 MM 25 MM	2109867-23	32
MAGNET RING BRACKET	2284172	1
SCREW PAN HEAD 6 MM 16 MM	2109873-25	2
CAP SCREW HEXAGON HEAD 10 MM X 1.5 X 35 MM	2109866-22	8
VACUUM GREASE MSDS# 8366442	2297760	1
SHOULDER WASHER NYLON	2308141-4	8
SCREW SET SCREW 10 MM 20 MM	2251294	4
RADIAL FOOT PADS	2310034	12

Field Supplied Tools

Disposable Gloves	Rags
Masking Tape	Rubbing Alcohol
Tongue Depressor	Allen Wrench 8 mm
Non Magnetic Tool Kit Metric/English	

3- MAGNET FLANGE CHECK

There may be issues with the Magnet Flange that will prevent the O-rings and gaskets that are between Magnet Flange and Alignment Ring from maintaining proper vacuum. It is estimated that 10% of all LCC magnets may have issues that will need to be addressed. To check for Magnet Flange consistency, place the Alignment Ring against the Magnet Flange. See Illustration 3-1. Refer to Appendix C if gaps or protrusions are discovered.

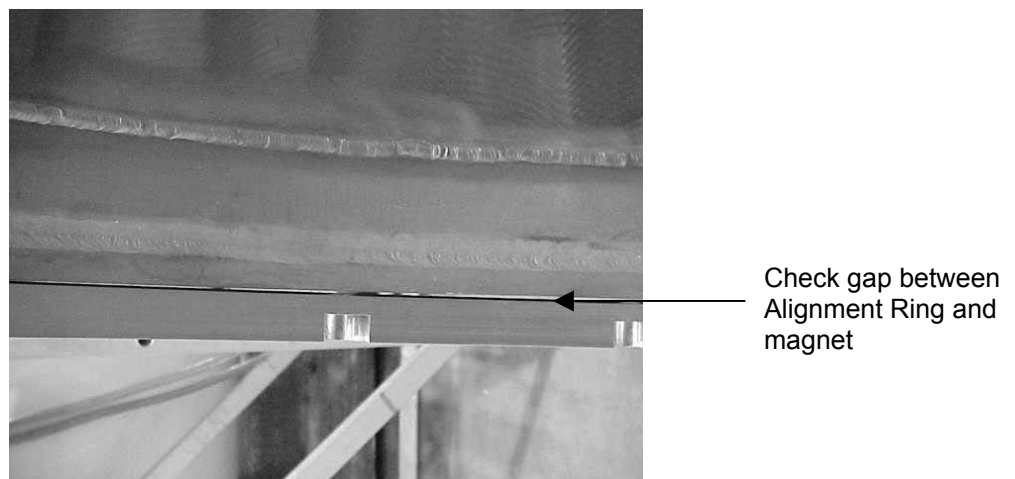


ALIGNMENT RING CHECK
ILLUSTRATION 3-1

Check for the following conditions:

- Protruding magnet weld
- Cavity in the flange
- Bolt pattern on the Magnet Flange does not match bolt pattern on the Alignment Ring
- Skewed Flange

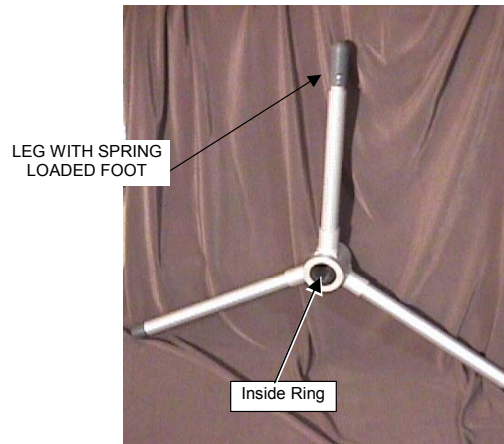
If any of the above conditions are noted, refer to Appendix C for the resolution needed.



GAP CHECK
ILLUSTRATION 3-2

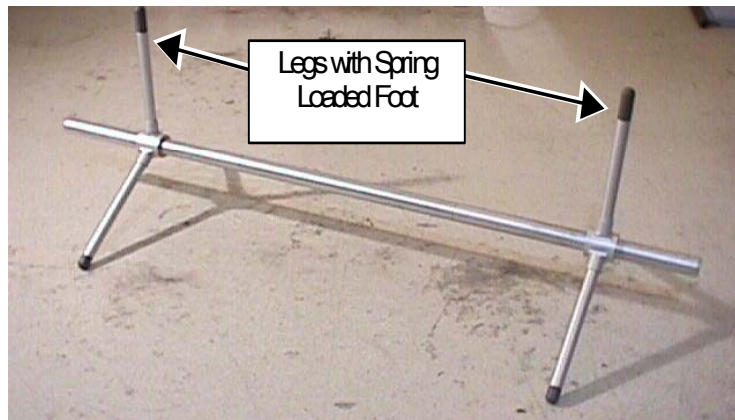
4- TOOL SETUP

1. Open the Twin Centering Tool case and assemble the parts for the two 3-legged spider assemblies as shown in Illustration 4-1. Note: each spider assembly gets one leg with Spring Loaded foot.



SPIDER ASSEMBLY
ILLUSTRATION 4-1

2. Wipe the center shaft and the inside ring of the Spider Assembly.
3. Assemble 3-piece center shaft and slide a spider assembly on at each end. The leg with the spring loaded foot should be pointed up.

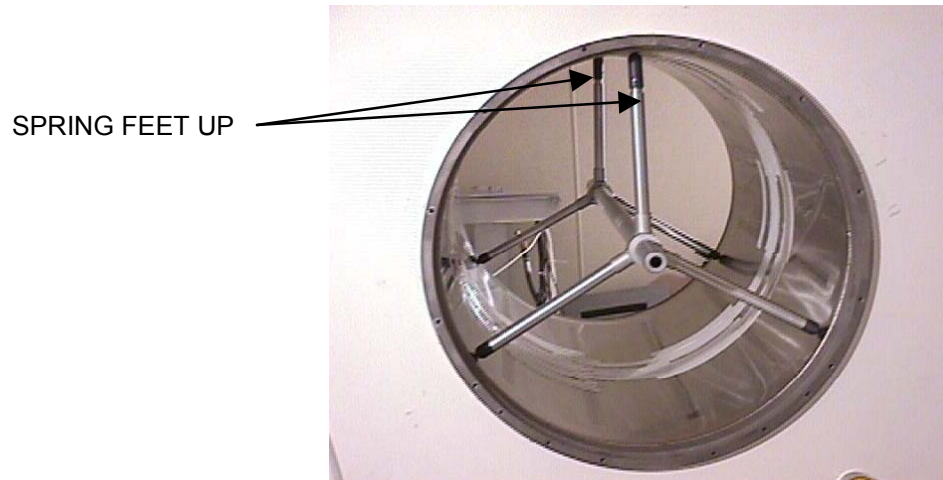


CENTER SHAFT WITH SPIDER ASSEMBLIES
ILLUSTRATION 4-2

4. Insert Center Shaft Assembly into magnet bore being careful not to drag across the passive shims. The legs with the spring loaded foot are pointed up. See Illustration 4-3.
5. After Center Shaft Assembly is in position, rotate the spring loaded foot of each Spider Assembly $\frac{1}{4}$ turn to activate the spring tension against the top of the magnet bore. *Hint:* Spider Assembly feet must maintain pressure against the magnet bore, outside of the passive shim area.

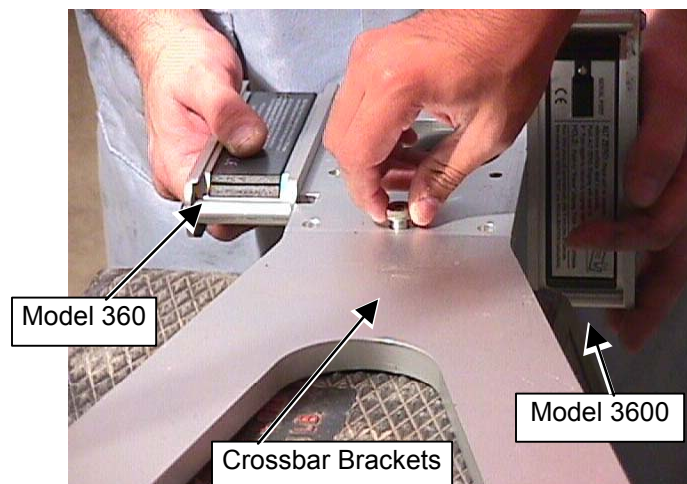
Note

It is important to place the spider assembly as close as possible to the passive shim location markers in the magnet bore and away from the weld seam.



CENTER SHAFT ASSEMBLY IN BORE
ILLUSTRATION 4-3

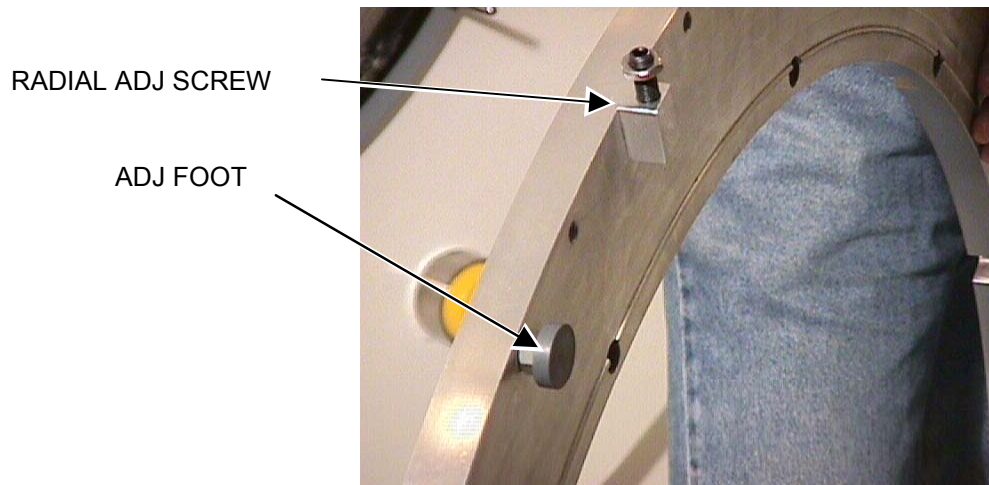
6. Connect the Digital Protractor Assembly to the crossbar brackets using the 6 allen-head capscrews and wrench provided in kit. Tighten snugly. See Illustration 4-4.



ASSEMBLE DIGITAL PROTRACTORS TO BRACKETS
ILLUSTRATION 4-4

7. Make sure the model 3600 protractor is mounted on the top and in line with the axial pipe and the model 360 protractor is mounted below and perpendicular to the axial pipe.
8. Wipe down both sides of the interface ring with alcohol.

9. Inspect magnet interface ring to ensure the adjustment feet and the radial screws are fully retracted. See Illustration 4-5. Note that in early production the foot screws are ferrous.



BACK OF INTERFACE RING – ADJUSTABLE SCREW/FOOT
ILLUSTRATION 4-5

10. Attach the Twin Centering Tool to the front of one of the enclosure interface rings using six 17 mm hex-head screws provided in kit (note: the front of the interface ring is opposite the side with the 4 round feet and the lower protractor readout faces out of the magnet, see Illustration 4-7). Tighten snugly. See Illustration 4-6.



TWIN CENTERING TOOL TO ENCLOSURE INTERFACE RING
ILLUSTRATION 4-6



READ DIRECTION OF LOWER PROTRACTOR
ILLUSTRATION 4-7

5- PREPARATION FOR INTERFACE RING

The Ring Alignment Tool is used to position the Vacuum Enclosure Interface Rings. There are some enclosure parts that must be installed before the interface rings.

1. The first enclosure part to be installed is the Wedge Ring that fits around the magnet's end ring. The Wedge Ring must sit flat against the magnet's end flange. It may be necessary to use non-ferrous tooling to remove any weld globs that inhibit the Wedge ring from sitting flat. See Illustration 5-1 showing removal of weld globs prior to installation of the Wedge Ring.

WARNING!

POSSIBLE PROPERTY DAMAGE AND IMAGE QUALITY ISSUES. THE WELD GLOBS ARE FERROUS MATERIALS AND MUST BE REMOVED FROM THE MAGNET ROOM TO PREVENT POTENTIAL WHITE PIXEL SOURCE.

WARNING!

POSSIBLE PERSONAL INJURY. THE WELD GLOBS WILL COME LOOSE AND COULD INJURE EYES. WEAR PERSONAL PROTECTIVE EQUIPMENT SAFETY GLASSES WHILE REMOVING GLOBS.



PREPARATION FOR WEDGE RING
ILLUSTRATION 5-1

2. Wipe down the magnet end flange and magnet ring with alcohol before installing wedge ring.

3. Install Wedge Ring. See Illustration 5-2. There is a notch in the Wedge Ring. The position of the notch is not relevant for the front of the magnet. There is a weld in the rear of the magnet that will allow you to position the notch over the weld.

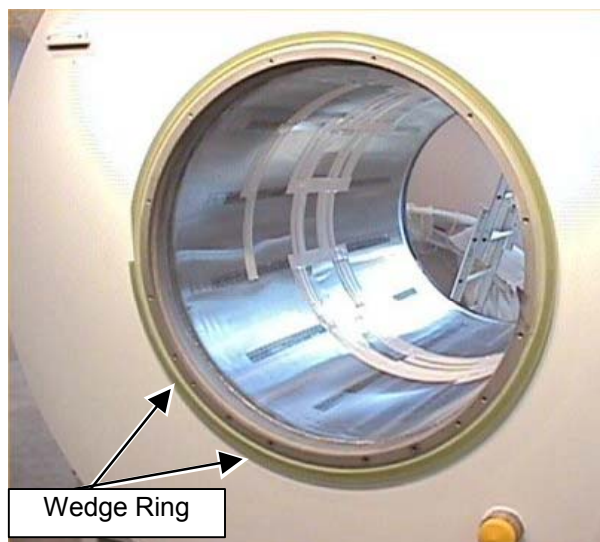
**WEDGE RING INSTALLED**

ILLUSTRATION 5-2

4. Using disposal gloves apply a thin film of vacuum grease (P/N 2297760) to the face of the magnet flange ring. See Illustration 5-3.

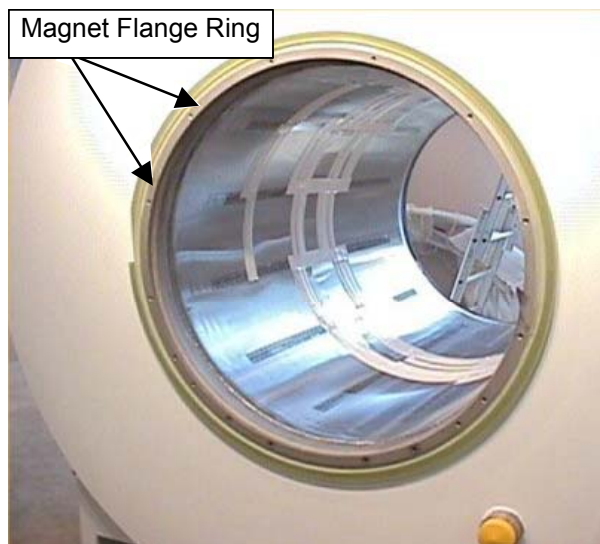
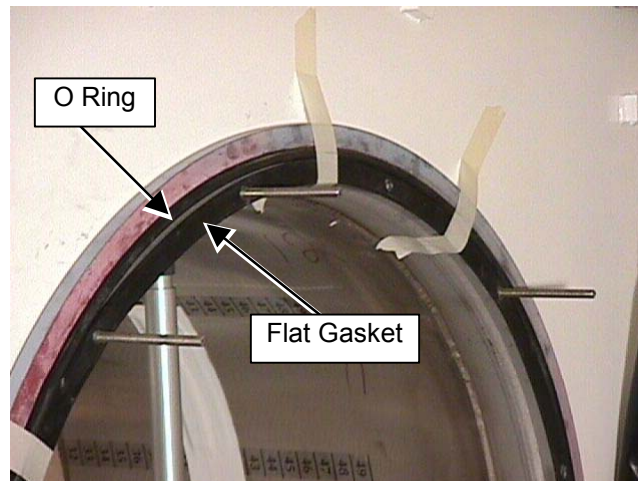
**MAGNET FLANGE-RING LOCATION**

ILLUSTRATION 5-3

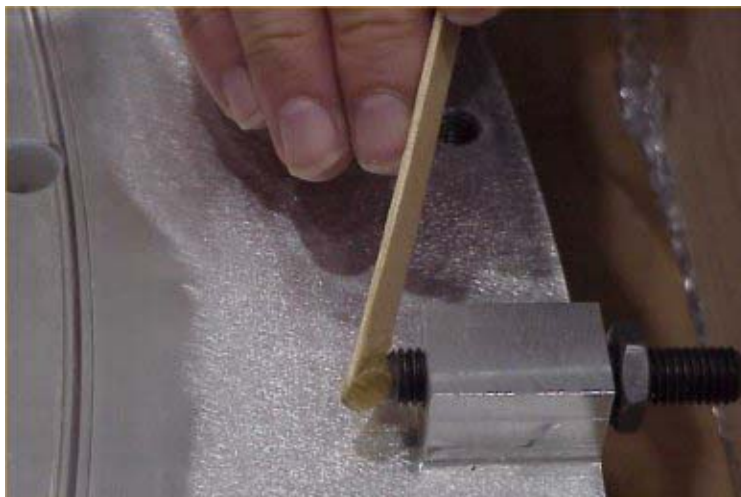
5. Apply vacuum grease to the 6 threaded studs that came with the enclosure and install into threaded holes on magnet ring. See Illustration 5-4. Place 3 studs, as shown, around top half of magnet end ring and 3 around the bottom half.
6. Install flat gasket onto the studs installed in previous step. See Illustration 5-4.
7. Using disposal gloves apply vacuum grease (P/N 2297760) liberally to the round O-ring (5/8" thick), part number 2310035. Then install O-ring around the magnet end ring. See Illustration 5-4.



TAPE FLAT GASKET IN PLACE

ILLUSTRATION 5-4

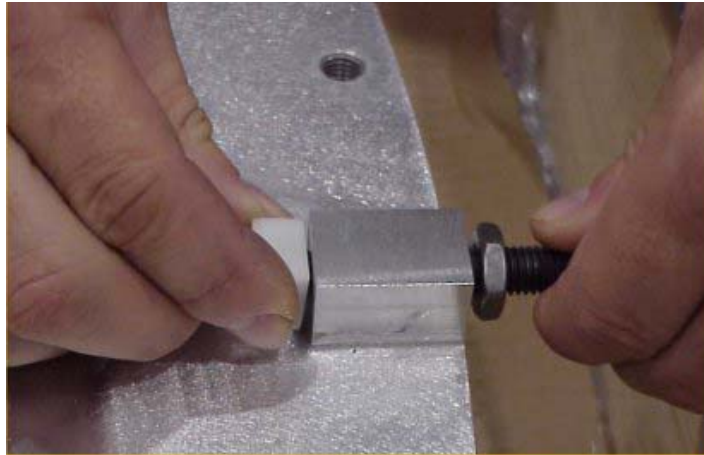
8. Use masking tape or similar to hold the O-ring into place. See Illustration 5-4.
9. Use a tongue depressor to place vacuum grease on the bottom of the radial screws. See Illustration 5-5.



APPLYING GREASE TO RADIAL SCREWS

ILLUSTRATION 5-5

10. Place the nylon spacers on the end of the radial screw. See Illustration 5-6.



PLACING THE NYLON SPACER

ILLUSTRATION 5-6

11. Place the Interface Ring with the Twin Centering Tool over the studs on magnet end ring. See Illustration 5-7.



BOLT INTERFACE RING TO FRONT OF MAGNET

ILLUSTRATION 5-7

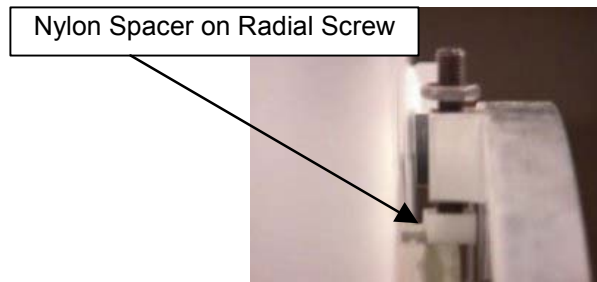
12. Remove the tape holding the flat gasket in place.

Note

In step 13, it may be necessary to use the nut with metal washer on the studs from the Gradient Insertion tool kit to start to tighten the interface ring to the magnet on both the front and rear rings of the magnet.

13. Put vacuum grease on threads before inserting the M10 x 25mm capscrews from the enclosure into the remaining holes around the interface ring. Just start the capscrew threads in the holes, do not tighten beyond that for now. After installing three screws, the threaded rods can be removed. They will be used when installing the rear Alignment Ring.

14. Inspect inside of magnet interface ring to ensure that the nylon spacers for radial screws are in place. See Illustration 5-8.



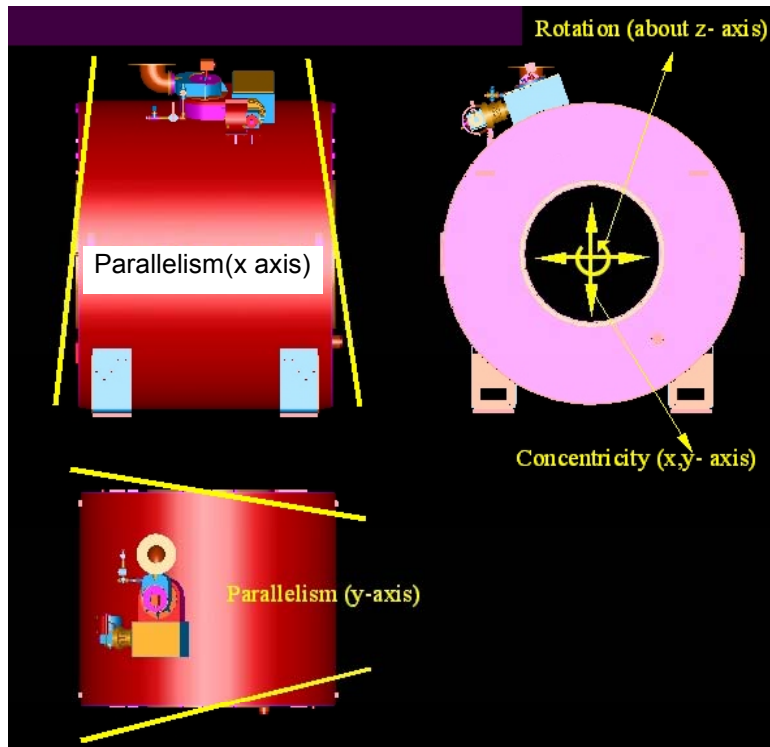
NYLON SPACER IN PLACE

ILLUSTRATION 5-8

15. Verify that the O-ring is in the proper place all around the wedge ring, otherwise damage to the O-ring could occur.

6- ALIGNMENT PROCEDURE

The alignment procedure requires patience and perseverance to be done correctly. The interface ring must be aligned in all physical planes to ensure proper alignment of the RF Coil to the TRM and vacuum system functionality. There are 4 alignment parameters. Concentricity alignment, rotational alignment, parallelism about x-axis and parallelism about y-axis, in the same order. See Illustration 6-1.



ALIGNMENT PARAMETERS
ILLUSTRATION 6-1

The alignments are completed for the front interface ring first and then repeated on the rear interface ring.

6-1 Rotational Alignment

The goal of the rotational alignment is to set the rotational position of the interface ring as close to zero degrees as possible (Note: The magnet should have been leveled at initial magnet installation). The Digital Protractor is used to set the reference.

1. Toggle ON/OFF switch on the Digital Protractor and check the reading. The arrow indicates which direction the interface ring must be moved. See Illustration 6-2.

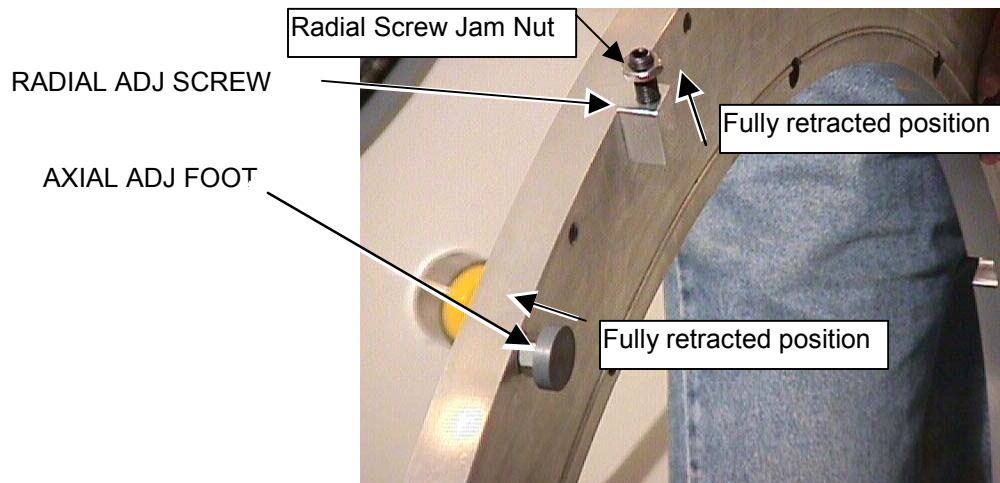


DIGITAL PROTRACTOR
ILLUSTRATION 6-2

2. Manually rotate the ring clockwise or counter clockwise to reach the goal of an alignment reading which is $\leq 0.5^\circ$.
3. Record final measurement.

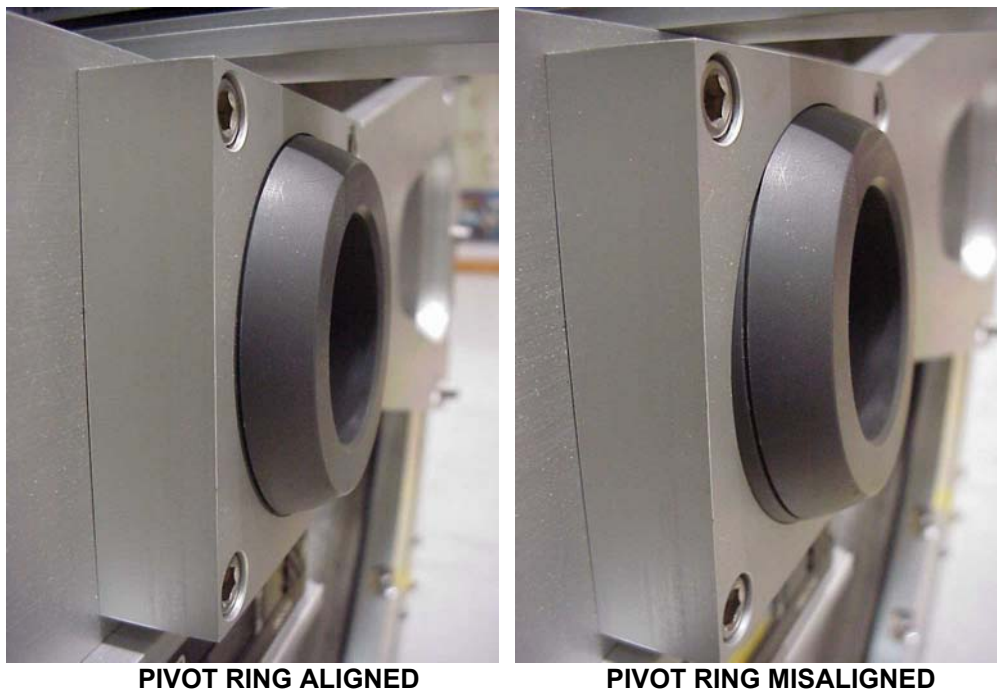
6-2 X-Y Plane Concentric Alignment

The goal of the X-Y plane concentric alignment is to position the interface ring such that the center shaft can be inserted through the pivot ring of the Digital Protractor assembly. See Illustration 6-2 for pivot ring. **Do Not attempt to force the shaft into the pivot ring**; this could damage the pivot ring. The interface ring must be positioned using the radial screws on the interface ring and then the center shaft is inserted **by hand** through the pivot ring. See Illustration 6-3 of the backside of the interface ring.



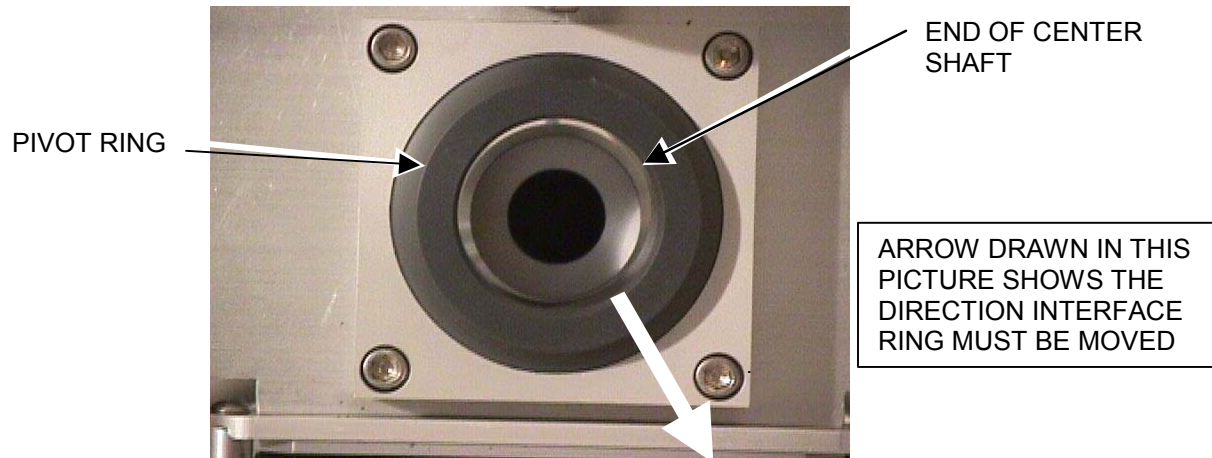
BACK OF INTERFACE RING – ADJUSTABLE SCREW/FOOT
ILLUSTRATION 6-3

1. Position the pivot ring so it is perpendicular to the center shaft assembly, make sure the groove is even or aligned to the face of the plate (see Illustration 6-4). Slide the center shaft close to the pivot ring.



PIVOT RING ALIGNED **PIVOT RING MISALIGNED**
ILLUSTRATION 6-4

2. Look through the hole of the pivot ring to see alignment of center shaft. See Illustration 6-5. Determine direction the interface ring must move to align the center shaft to the pivot ring opening.



VIEW OF CENTER SHAFT THROUGH PIVOT RING
ILLUSTRATION 6-5

3. Use Allen wrench to turn the appropriate radial adjustment screws to move the interface ring into alignment with the center shaft.
4. Continue to make adjustments using the radial adjustment screws until the shaft can be pushed through the pivot ring using only the force of your hand. See Illustration 6-6.



CENTER SHAFT THROUGH PIVOT RING
ILLUSTRATION 6-6

Note

Due to the stack up of tolerances of the passive shim in the magnet bore; it may be necessary to shim the legs of the spider assembly. The 8mm shims work well for this. Then repeat steps 1 – 4 till center shaft moves through pivot ring.

5. Place all radial fingers snug to the wedge.

6. Evenly tighten 16 capscrews a $\frac{1}{2}$ turn using an alternating pattern maintaining a gap of 8mm between the wedge ring and the magnet interface ring. For example picture the magnet ring as the face of a clock, tighten the capscrew at ten o'clock position, followed by the capscrew at 4 o'clock, followed by the capscrew at 7 o'clock ...

Note

Use a 8mm allen wrench to check the gap between the wedge ring and the magnet interface ring.

6-3 X & Y-Axis Alignment

The goal of the Y-axis alignment is to adjust the axial position of the interface ring as close to zero degrees as possible using the Axial Digital Protractor.

Note

Start with the center shaft slid out of the Pivot Ring completely.

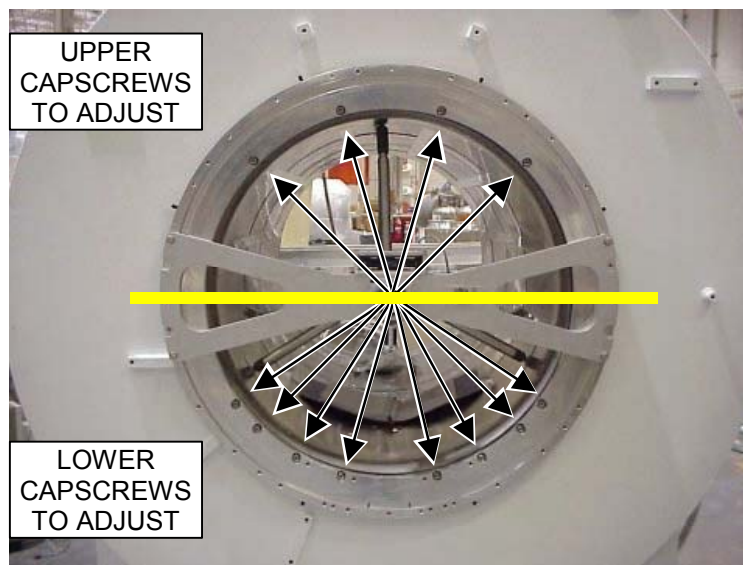
1. Toggle ON/OFF switch on the Axial Digital Protractor and check the reading. The arrow indicates which direction the interface ring must be moved. See Illustration 6-7.



AXIAL DIGITAL PROTRACTOR
ILLUSTRATION 6-7

6-3-1 Y-Axis Alignment

1. Using the directional arrow on the digital protractor, adjust the capscrews on the lower or upper half of the magnet interface ring to reach the goal alignment reading of $\leq 0.5^\circ$. See Illustration 6-8, the solid horizontal line divides the ring into the upper and lower halves.

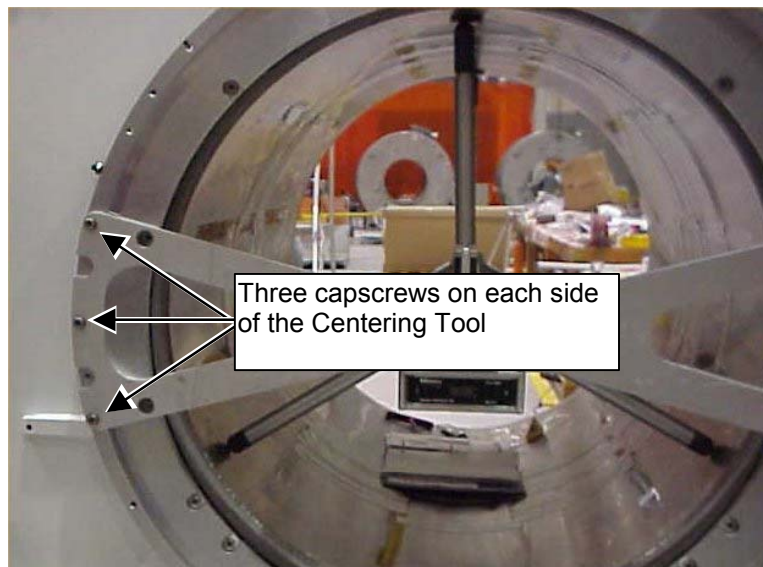


CAPSCREW ADJUSTMENT LOCATIONS
ILLUSTRATION 6-8

2. Record measurement.

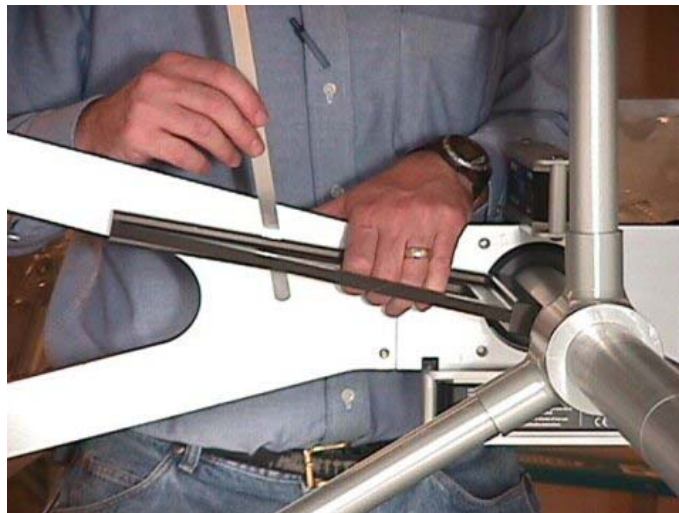
6-3-2 X-Axis Alignment

1. Loosen the six (6) screws that hold the centering tool on the interface ring. This is done to remove any distortion caused by over tightening the centering tool. See Illustration 6-9.



SCREW LOCATION
ILLUSTRATION 6-9

2. Use the right angle T-square at the shaft and use the 0.25 mm gauge to make sure the gap does not exceed the gauge. To make the adjustment tighten the capscrews on the left or right side to adjust the gap which aligns the X-axis. See Illustration 6-10.



RIGHT ANGLE T-SQUARE MEASURING X-AXIS ALIGNMENT
ILLUSTRATION 6-10

3. After the axial alignment has met the alignment goal, extend all adjustment feet only finger tight to just make contact with magnet front face, and then lock with nut.
4. Tighten the radial screws evenly and then lock in place with the jam nut. See Illustration 5-9 for examples of radial screw locations.

6-4 Securing Adjustments

1. Check to ensure that the radial screws and axial feet are locked down.
2. Write down the final alignment numbers from the digital protractors.
3. Remove the 6 capscrews that hold the Centering Tool Assembly and install on the rear interface ring.

6-5 Rear Interface Ring Alignment

Repeat sections 4 and 5 for the Rear Interface Ring installation.

APPENDIX A – DIGITAL PROTRACTOR TESTING

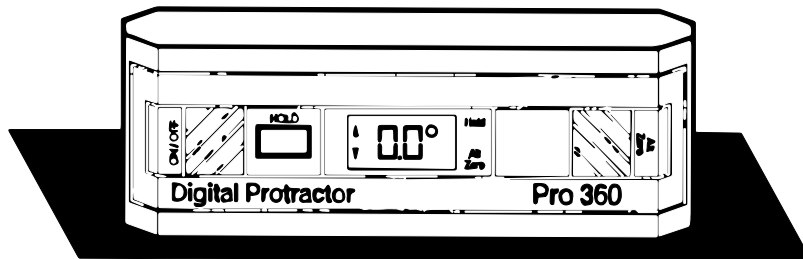
A-1 Recalibration

Like all measurement instruments the Pro 360/3600 must periodically be monitored for accuracy. The Pro 360/3600 should be tested daily with the simple procedure outlined below. If it is found to be out of calibration, it is easily recalibrated on site using Superset® - an eight-step procedure that realigns the unit through its entire 360° range and takes just a couple of minutes to perform. And Superset® doesn't require any special fixtures, tools, or expertise.

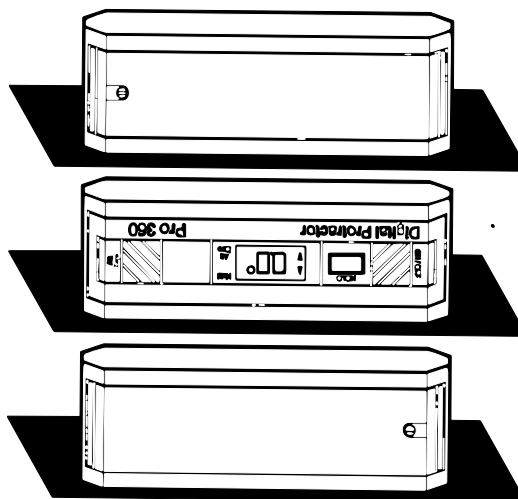
Accuracy Test

Perform this simple test each day before using the Pro 360/3600. Also perform it anytime the Pro 360/3600 has been dropped or is being used in an environment that varies more than 5° C (9° F) from the environment in which it was last calibrated. *If your Pro 360/3600 fails this test, you must recalibrate it using Superset®, before using the unit further to measure angles.*

1. Position the Pro 360/3600 with the display facing you on a clean, flat horizontal surface. It doesn't have to be exactly level. Wait 10 seconds so the unit is completely settled and note the angle on the display.



2. Rotate the unit end-for-end so the display is facing away from you. Be sure to set the Pro 360/3600 in exactly the same spot, and wait 10 seconds before reading the angle that's displayed.



3. Now roll the unit toward you so that the display is facing you, but the lettering on face of the unit is upside down. Wait 10 seconds, and note the angle on the display.
4. Finally, rotate the unit end-for-end so the display is facing away from you (the lettering should still be upside-down). Wait 10 seconds and note the angle on the display.

5. If any of the four measurements varies from any other by more than 0.1° , you must recalibrate (Superset[®]) your unit before using it further.

Superset[®]

Superset[®] recalibrates the Pro 360/3600 through its entire 360° range by electronically recording four horizontal and four vertical settings. It should be performed whenever the accuracy test shows a discrepancy of 0.1° or more.

How to Perform Superset[®]

1. Turn on the Pro 360/3600 and place it on a flat surface. You can use any horizontal surface within 10° of level and any vertical surface within 10° of plumb to perform Superset[®]. You must use the same surfaces throughout the entire process.

Note

Each time you reposition the Pro 360/3600 during Superset[®], wait a minimum of 10 seconds before pressing the **[HOLD]** button to advance to the next step.

2. Press and hold the **[HOLD]** and **[ALT ZERO]** buttons simultaneously. Keep them depressed for approximately three seconds.
3. Release the buttons when the symbol "SUP" appears. A "0" within flashing brackets will then appear. These brackets are composed of four horizontal and four vertical segments.



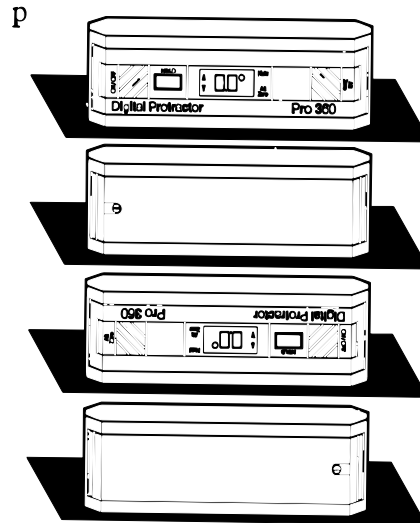
"0" within flashing brackets

Note

As you proceed through the next eight steps (4 – 12) of Superset[®], a new segment will hold steady after you complete each step.

Superset® - Horizontal Settings

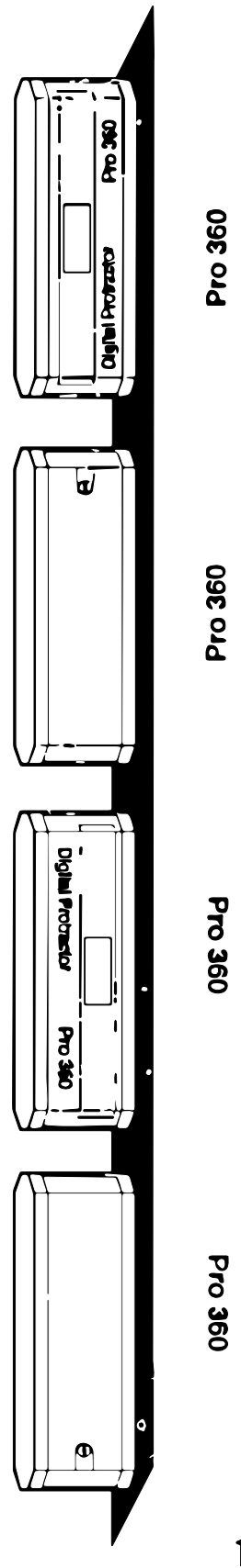
4. The unit faces you and the white lettering on face is right-side up. Align with an edge or line - wait 10 seconds. Press [HOLD] button until 1 appears.
5. Rotate the unit so it faces away from you, the lettering should still be right-side up. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 2 appears.



6. Roll the unit so it faces you, the lettering should now be upside-down. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 3 appears.
7. Rotate the unit so it faces away from you, the lettering should still be upside-down. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 4 appears.

Superset® - Vertical Settings

8. Place the unit against vertical surface so it faces you, the lettering on the face (Pro 360, etc.) should read from bottom-to-top. Align with an edge or line - wait 10 seconds. Press [HOLD] button until 5 appears.



9. Roll the unit so it faces away from you, the lettering should still read from bottom-to-top. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 6 appears.
10. Rotate the unit end-for-end so it faces you, the lettering should now read top-to-bottom. Align with same edge or line - wait 10 seconds. Press [HOLD] button until 7 appears.
11. Roll the unit so it faces away from you, the lettering should still read top-to-bottom. Align with same edge or line - wait 10 seconds. Press [HOLD] button. An 8 will very briefly appear, followed immediately by regular angle measuring.
12. Your Pro 360/3600 has been Superset[®] back to manufacturer's spec's.
Canceling Superset[®]
You may cancel Superset[®] at any time during the process by turning the unit off.

APPENDIX B – TWIN CENTERING TOOL KIT

The Twin Centering Tool Kit is packed in the case as follows.



TOP LEVEL OF KIT
ILLUSTRATION B-1

Item No.	Description
1.	Pivot Ring and Digital Protractors
2.	Instruction Manuals for Digital Protractors
3.	Right Angle
4.	.25 mm Feeler Gauge
5.	Leg for Spider Assembly
6.	Center Shaft
7.	Leg for Spider Assembly
8.	Part Box containing: 6 Hex head screws, 17 MM 1 Allen wrench 6 Bolts 6 Washers
9.	Leg for Spider Assembly with Spring loaded Foot

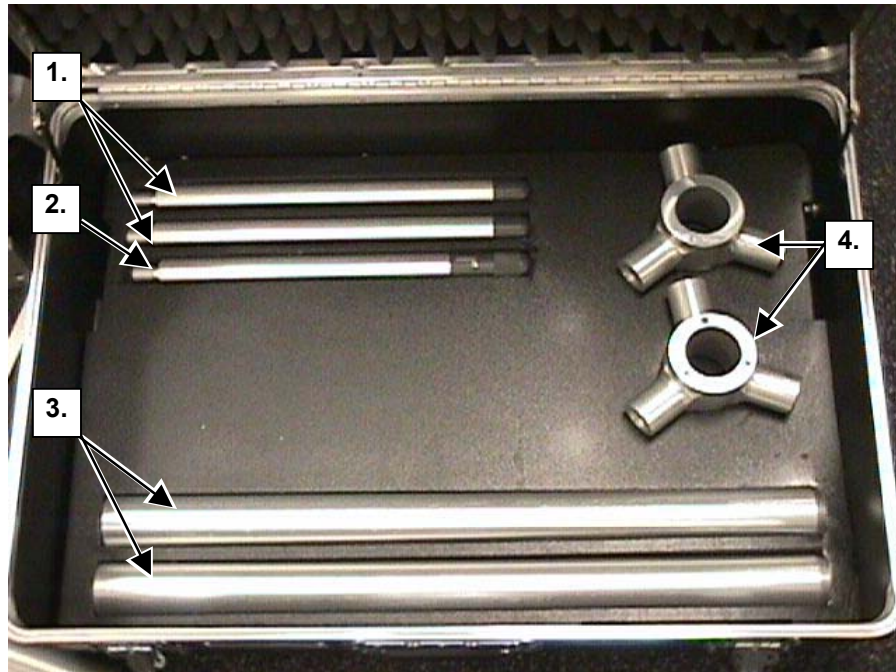


SECOND LEVEL OF KIT
ILLUSTRATION B-2



THIRD LEVEL OF KIT
ILLUSTRATION B-2

Item No.	Description
1.	Cross Bar Assembly
2.	Cross Bar Assembly



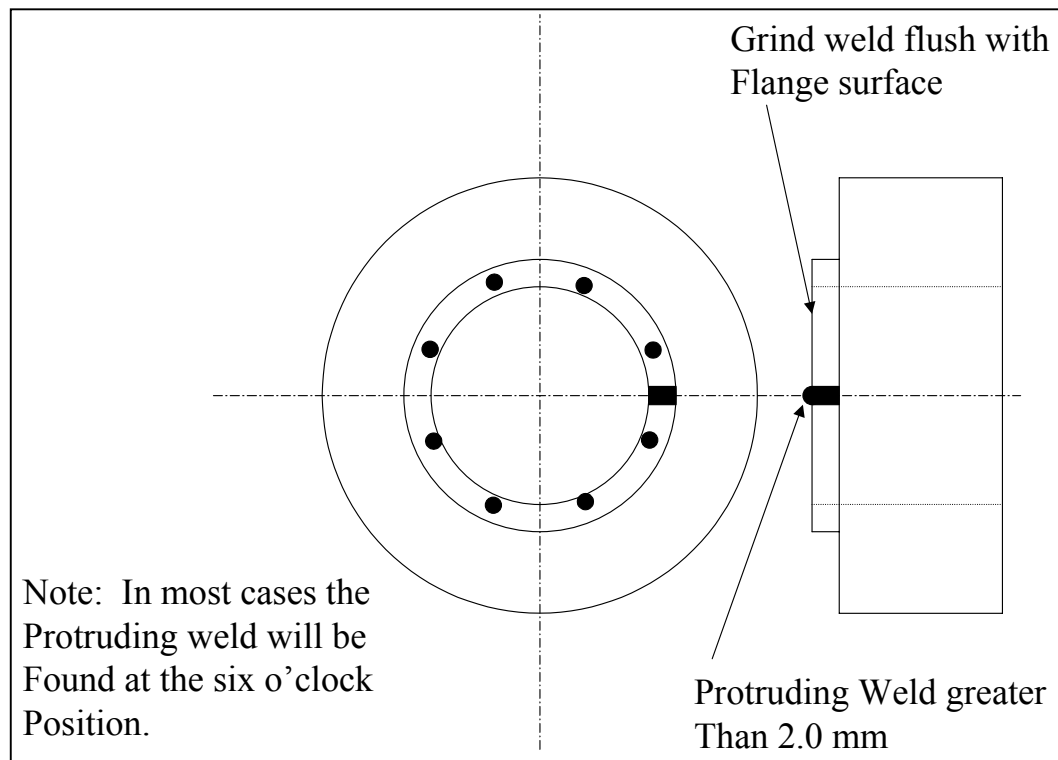
BOTTOM LEVEL OF KIT
ILLUSTRATION B-4

Item No.	Description
1.	Legs for Spider Assembly
2.	Leg for Spider Assembly with Spring loaded Foot
3.	Center Shaft Ends
4.	Spider Assemblies

APPENDIX C – MAGNET FLANGE ISSUES

If any of the following conditions are observed additional actions must be performed to allow proper fit between the Alignment Ring and the magnet flange. Worst case is a protruding magnet weld. The magnet must be ramped down for this situation.

C-1 Protruding Magnet Weld



If the Protruding weld is greater than 2.0mm, you must ramp the magnet down and grind off the weld.

IMPORTANT

Inform the customer that the Upgrade will take longer than expected.

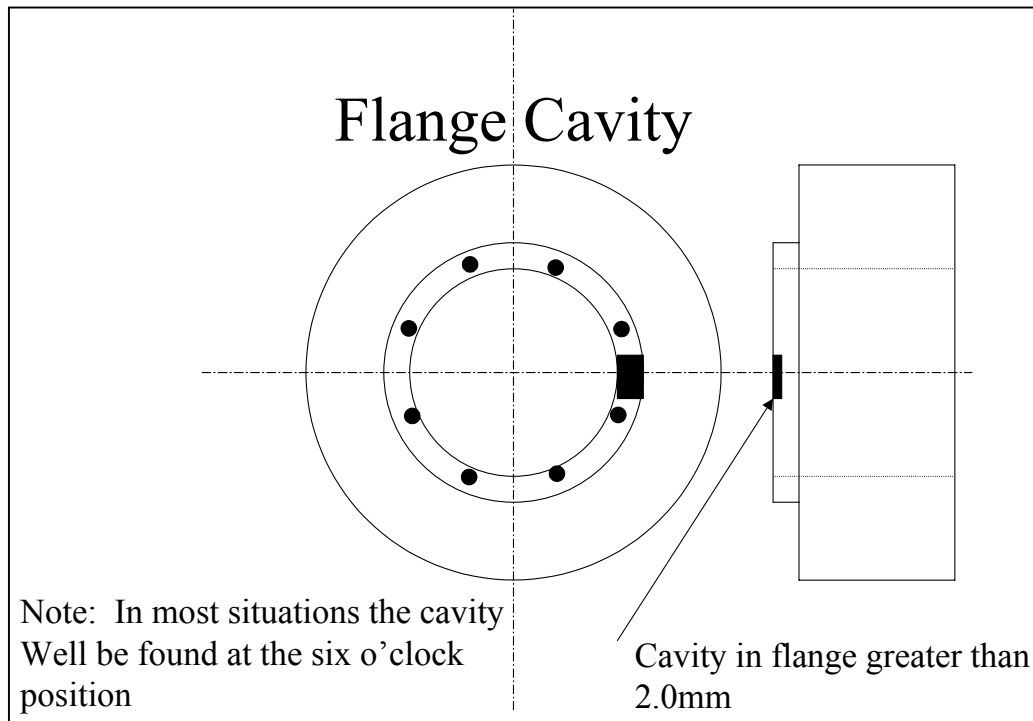
Tools needed:

- Ramp supply, Shim supply and associated equipment needed to ramp magnet.
- Die Grinder/Electric Drill
- Abrasive media for above selected tool
- Shop Vacuum or similar
- Drop Cloth or similar
- Tape
- Rags
- Alcohol

Procedure:

1. Ramp magnet down using standard procedure.
2. Remove the gradient coil from the magnet.
3. Use Tape and Drop Cloth, cover the area around the protruding weld of the magnet.
4. Prior to grinding turn the Shop Vac on to capture the debris from the grinding action.
5. Grind off the weld so the magnet flange surface is flush within .5mm.
6. Remove tape and drop cloth, then clean the magnet with rags and alcohol.
7. Install the TRM gradient coil.
8. Ramp the magnet.
9. Complete the Upgrade.

C-2 Cavity in Magnet Flange



If any Cavity on the Magnet Flange is greater than 2mm, it must be filled in with RTV.

Tools Needed:

- RTV
- Level or Straight Edge

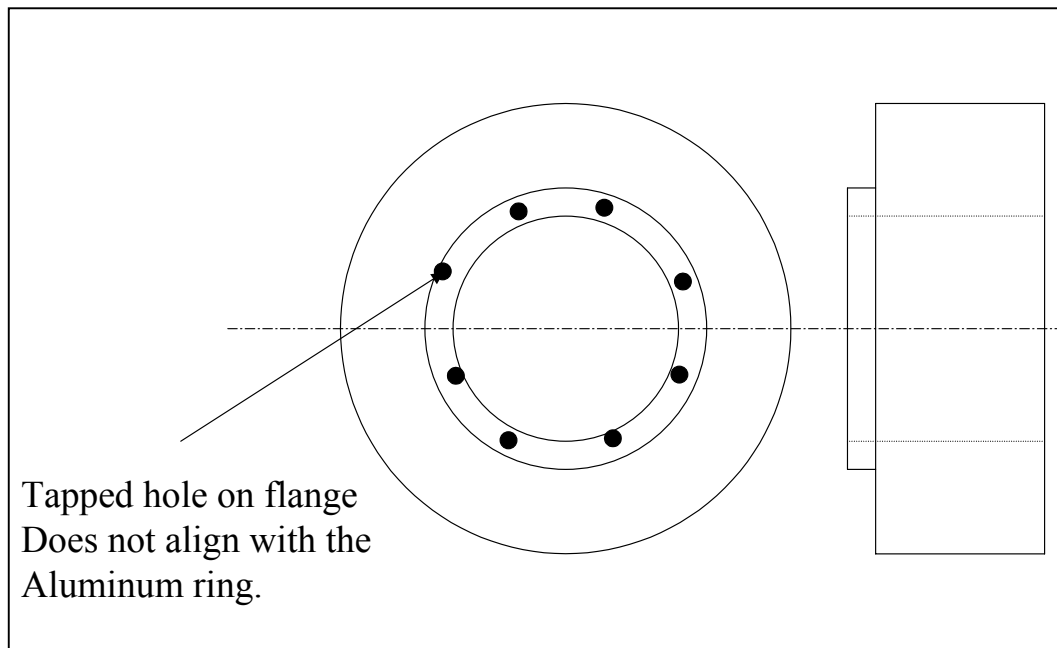
Procedure:

1. Using a level determine the amount of RTV needed to fill in the cavity.
2. Fill in the cavity with RTV using a straight edge.

IMPORTANT

The RTV must be supported to prevent it from being pulled into the vacuum space.

C-3 Miss-aligned Hole in Magnet Flange



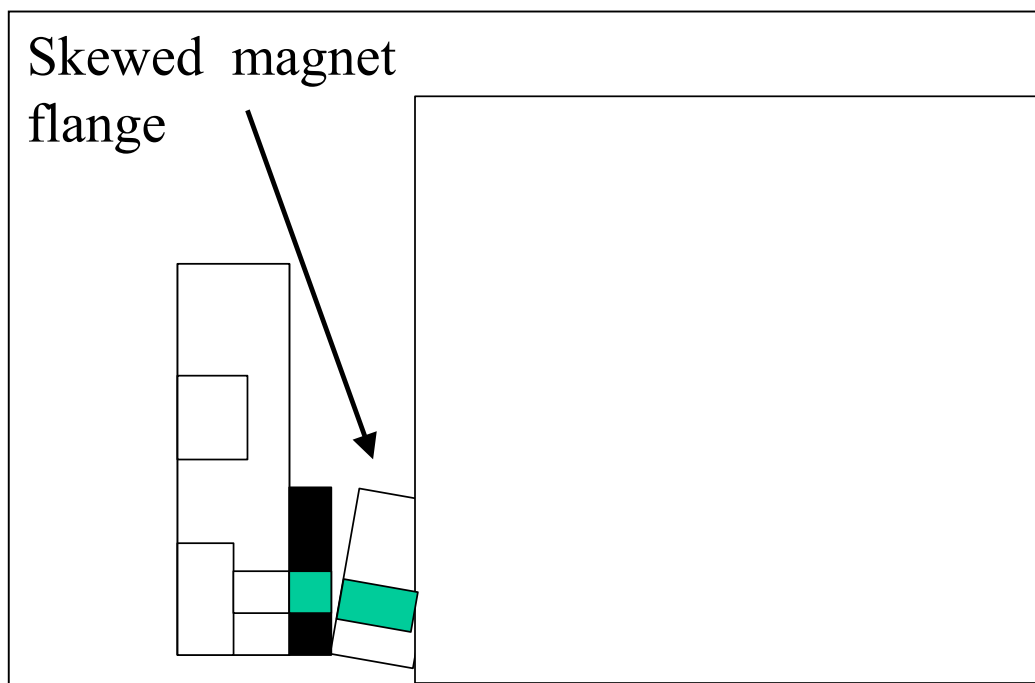
Tools needed:

- RTV

Procedure

1. Remove the plastic bushing in the Aluminum Alignment Ring. Try to install the bolt with the bushing removed. If the bolt can be threaded into the magnet, use Electrical tape to isolate the bolt from the Alignment Ring.
2. One bolt can be left out per flange, provided that the bolt does not support the TRM coil. Downside to this method will be a vacuum leak. The open bolthole must be filled with RTV to avoid a vacuum leak.
3. The Aluminum Alignment Ring may need to be drilled out. Determine new diameter of hole and take alignment ring to a machine shop to be drilled out. Use a non-conductive washer to isolate the Alignment Ring from the bolt.

C4 Skewed Magnet Flange



Tools Needed:

- RTV
- Level or Straight Edge
- Square

Procedure:

1. Using a straight edge or square, determine amount and location of RTV needed on the magnet to produce a flat surface.
2. Apply RTV to the magnet flange to create a parallel surface for mounting of the Alignment Ring.